

data transmission will be wireless (either radio or, eventually, radiated laser light), the connections between traffic hubs and nodes could always be interrupted when planetary rotation or orbital motion takes a transmitting entity out of line-of-sight of the receiving entity. This is a big challenge for realising a stable backbone.

Light takes over a second to travel to the moon and from Mars it takes anywhere from 3 to 15 minutes to reach the earth. These sorts of long delays would doom a space connection using the standard protocols that govern Internet communication, because they require the sending node to get a confirmation from the recipient node that each packet has been received correctly.

Applications and standards

The potential applications of the interplanetary Internet will just not be limited to the management of space missions. We, who surf the Internet today and use websites in extreme locations such as Antarctica, may some day be able to communicate to the Web or FTP servers on Martian micro satellites to request data directly from Mars. Eventually, there will be Web servers on the International Space Station, on all the planets and other regions of the solar system. We may even dream of webcams in space.

Security application and standard are a big challenge in IPN. In the terrestrial Internet, security has got a high priority due to the tremendous growth in ubiquitous connectivity and electronic commerce. Terrestrial Internet security is realised by IP layer security, secure sockets layer/transport layer security, and secure electronic mail using S/MIME or pretty good privacy. These security mechanisms, while operating at different layers of the Internet protocol suite, provide the required security service like confidentiality, integrity, authentication and non-repudiation.

IPN requires exactly these levels of security services. It differs from the terrestrial Internet in that the bandwidth

available is small and connectivity is not always guaranteed between IPN entities. As a result, the security mechanisms implemented in the IPN cannot assume that they can interact with their peers in an online and/or interactive fashion. The entities on the IPN have to operate in the IPN's pony-express model of operation—analogue to the way in which e-mail security operates. E-mail operates in a non-interactive mode and as such the security technologies developed for e-mail are directly adaptable in the IPN.

Since 1980s the Consultative Committee for Space Data Systems (CCSDS) has been standardising space communication techniques. The standards

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of CCSDS will be the basis for IPN communication standards. The wireless standards, coding standards, link standards, network standards, security standards, end-to-end delivery standards and file transfer standards of CCSDS will be the source and references for IPN standards.

Deployment strategies

Terrestrial Internet did not happen overnight. It took about 20 years to become a practical system of networking. Deep space network and standards are the source of beginning for the IPN.

The National Aeronautics and Space Administration (NASA) currently communicates with interplanetary and earth-orbiting missions using its deep space network. The deep space network works on line-of-sight communications. Thus rovers on Mars have to wait until they are back in the line of

sight with the earth before sending a message home. To receive the message, the deep space network dishes must be pointed in the right direction at the right time, or the signal will be lost forever. The Mars exploration rovers will be equipped with CCSDS protocols that will eventually make up the IPN infrastructure.

NASA in Nov.-Dec. 2008 had announced successful space tests of its new purpose-designed interplanetary communications networking protocol known as delay-/disruption-tolerant networking (DTN). A DTN is designed such that temporary or intermittent communication problems, limitations and anomalies have the least possible adverse impact. There are several aspects to the effective design of a DTN, including fault-tolerant technique, graceful degradation, quick recovery and minimal latency.

"This is the first step in creating a totally new space communications capability—an interplanetary Internet," said Adrian Hooke, at NASA headquarters in Washington.

DTN differs significantly from the existing Internet's TCP/IP. Unlike TCP/IP, it does not assume a continuous end-to-end connection. If a destination path cannot be found, data packets are not discarded. Each node keeps the information as long as necessary until it can communicate safely with another node. This store-and-forward method means that the information does not get lost when no immediate path to the destination exists. Eventually, the information almost always gets through.

NASA is already studying designs for a Mars network of multiple orbiting relay satellites. This will eventually be the first segment of the interplanetary backbone. The interplanetary Internet will possibly wire us to Mars within the next decade and to other planets in the decades to follow. It is a natural evolution of the Internet and big challenge of the future! ●

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